APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

SOFR / LMM Term Rate Model

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Owner	Dr. Yuki Tanaka (Quant Researcher)
Validator	Dr. Samuel Achebe (Model Validation Officer)
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Regulatory Framework: ARRC SOFR Transition Guidance, ISDA 2020 Fallback Protocol

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1 Model Overview

1.1 Purpose

Following LIBOR cessation on 2023-06-30, this model governs SOFR compounding-in-arrears rate computation, Term SOFR CME reference rate usage, and LIBOR fallback mechanics per the ISDA 2020 protocol. The LMM extension provides cap/floor pricing in the SOFR framework.

1.2 Business Use

- **Loan and swap pricing:** All floating-rate instruments now reference SOFR.
- **Fallback processing:** Legacy LIBOR contracts converted per ISDA waterfall.
- **Cap/floor pricing:** LMM used for client hedging products.
- **FRTB delta-IR:** SOFR curve sensitivities replace LIBOR curve under FRTB.

2 Theoretical Basis

2.1 SOFR Compounding In Arrears

For a SOFR-referencing floating period $[T_s, T_e]$:

$$R_{\text{comp}} = \left(\prod_{i=0}^{N-1} \left(1 + \frac{\text{SOFR}_i \cdot d_i}{360} \right) - 1 \right) \times \frac{360}{T_e - T_s}$$

where SOFR_i is the daily fixing, d_i is the day count (ACT/360), and N is the number of business days in the period.

Lookback mechanics: 5-business-day lookback (observation date = settlement date -5 bd) allows rate to be known 2 days before payment date.

2.2 LIBOR Fallback Waterfall (ISDA 2020)

Upon LIBOR cessation:

1. **Primary:** Term SOFR + spread adjustment (ISDA-prescribed static spread).
2. **Fallback 1:** Compounded SOFR in arrears + spread.
3. **Fallback 2:** Central bank rate + spread.
4. **Fallback 3:** Last published LIBOR rate (short-term disruptions only).

ISDA-prescribed USD LIBOR spread adjustments (5-year historical median):

Tenor	Spread Adjustment
1M LIBOR → SOFR	+11.448 bps
3M LIBOR → SOFR	+26.161 bps
6M LIBOR → SOFR	+42.826 bps

2.3 SOFR LMM

The SOFR LMM models the joint evolution of SOFR forward rates $F_j(t)$ for tenor periods $[T_j, T_{j+1}]$:

$$dF_j(t) = \mu_j(t)\, F_j(t)\, dt + \sigma_j\, F_j(t)\, dW_j(t)$$

Under the spot measure:

$$\mu_j = \sum_{k=j+1}^{n} \frac{\rho_{jk}\, \sigma_j\, \sigma_k\, F_k(t)\, \tau_k}{1 + F_k(t)\, \tau_k}$$

Caplet pricing (Black formula under LMM):

$$\text{Caplet}(T_j, K) = P(0, T_{j+1})\, \tau \big[F_j(0)\, N(d_1) - K\, N(d_2)\big]$$
$$d_1 = \frac{\ln(F_j/K) + \frac{1}{2}\sigma_j^2 T_j}{\sigma_j \sqrt{T_j}}, \qquad d_2 = d_1 - \sigma_j \sqrt{T_j}$$

3 Mathematical Specification

Parameter	Value
Tenors modelled	1M, 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y
LMM vol calibration	ATM SOFR caplet market quotes
Lookback convention	5 business days (ARRC recommendation)
Day count	ACT/360
Business day convention	Modified Following (NY)

4 Implementation

Code location: `infrastructure/treasury/ftp.py` — SOFR-based FTP curve uses compounded SOFR approximation via swap curve.

Calibration: Vol surface calibrated quarterly to CME SOFR caps. Spread adjustments hard-coded per ISDA published values.

5 Validation

SOFR compounding accuracy: Replication test: compound 20 days of SOFR fixings; confirm within \$0.01 of analytic formula.

Spread adjustment: ISDA-prescribed spreads confirmed against published ISDA supplement.

LMM caplet: Benchmark against Black caplet at ATM; within 1 bp implied vol.

6 Model Limitations

1. **SOFR-Fed Funds basis:** SOFR/Fed Funds basis not explicitly modelled; can diverge by 5–20 bps during month-end/quarter-end.

2. **Term SOFR restrictions:** ARRC prohibits Term SOFR in financial derivatives not hedging loans; compliance team must approve each Term SOFR usage.
3. **LMM vol surface:** Calibrated using pre-LIBOR vol surface as proxy during 2022–2023 transition; SOFR-specific vol history now available.
4. **Backward-looking risk:** SOFR in arrears creates P&L uncertainty during the accrual period; risk systems must account for daily revaluation.

7 Open Findings

ID	Severity	Description	Status
LMM-F1	**Major**	SOFR-Fed Funds basis risk not modelled; EOM/EOM spikes can cause 15–20 bps P&L mis-estimation.	Open
LMM-F2	**Major**	Term SOFR use-case documentation incomplete; ARRC pre-approval not recorded for 3 existing client contracts.	Open
LMM-F3	**Minor**	LMM vol calibration uses pre-LIBOR proxy surface; SOFR-native vol history sufficient for full recalibration since 2024-Q1.	Open